Buoyancy

We know that gravity pulls objects downwards. The force is known as weight.

But underwater there's another, upward force, called buoyancy.

If the weight of an object is greater than the buoyant force, the object will sink.

If the buoyant force is greater than the weight, the object will float.

The amount of buoyant force is determined by the weight of water an object moves out of the way - known as the displaced water.

Buoyant Force:
An upward acting force exerted by a fluid
Opposes an object's weight

The greater the weight of the displaced water, the greater the buoyant force.

Whether an object floats or sinks depends on the density of the object in comparison to the density of the fluid it displaces.

Density = amount of mass in a set volume

If the density of the object is greater than that of the water it's displaced, its weight will be greater too and it will sink.

If it's less dense, it will float.

The effects of changing the density can be seen in the ease with which people float on the salty water of the Dead Sea.

The salt increases the density of the water, so the buoyant force is much larger.

Buoyancy determined by:
Volume **of displaced water and object**
Density **of displaced water and object**

These principles help explain why heavy metal ships float.

Firstly, they're big - they displace a huge amount of water, creating a very strong buoyant force.
And secondly, they are hollow - combined, the metal shell and air in the hull are less dense than water, so the weight is less than the buoyant force.
If ships were made of solid metal, they would sink!